

# Veronica Kelsey

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## Work and Education

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- 2021 - 2027 **Heilbronn Research Fellow** The University of Manchester
- 2018 - 2021 **PhD Mathematics** University of St Andrews  
Supervisors: Prof Colva M. Roney-Dougal and Dr Martyn Quick  
Thesis: Base size and generating graphs of primitive permutation groups
- 2014 - 2018 **MMath Mathematics** The University of Manchester  
Dissertation: Mathieu groups and Chamber Graphs

## Peer Reviewed Publications

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- [1] **The excess zero graph of a Coxeter group**  
with S.B. Hart and P.J. Rowley - J. Group Theory (To appear)
- [2] **Root cycles in Coxeter groups**  
with S.B. Hart and P.J. Rowley - J. Group Theory (2025)
- [3] **The relational complexity of linear groups acting on subspaces**  
with S.D. Freedman and C.M. Roney Dougal - J. Group Theory (2025)
- [4] **A note on the rank 5 polytopes of  $M_{24}$**   
with R. Nicolaides and P.J. Rowley - Innov. Incidence Geom. (2024)
- [5] **On the rank 5 polytopes of the Higman-Sims simple group**  
with R. Nicolaides and P.J. Rowley - Innov. Incidence Geom. (2022)
- [6] **On relational complexity and base size of finite primitive groups**  
with C.M. Roney-Dougal - Pacific J. Math. (2022)
- [7] **Maximal cocliques in the generating graphs of alternating and symmetric groups**  
with C.M. Roney-Dougal - Comb. Theory (2022)
- [8] **Generating finite Coxeter groups with elements of the same order**  
with S.B. Hart and P.J. Rowley - Turkish J. Math. (2021)
- [9] **Chamber graphs of minimal parabolic sporadic geometries**  
with P.J. Rowley - Innov Incidence Geom. (2020)
- [10] **A note on involution centralizers in black box groups**  
with P.J. Rowley - J. Group Theory (2020)
- [11] **Chamber graphs of some geometries that are almost buildings**  
with P.J. Rowley - Innov. Incidence Geom. (2019)
- [12]  **$M_{24}$  orbits of octad triples**  
with P.J. Rowley - Graphs Combin. (2018)

## Preprints and Papers in Preparation

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- [1]  **$\mathbb{P}(q)$ -Groupoids of Conway Type**  
with P.J. Rowley - submitted - arXiv:2410.22202
- [2] **Matroids, IBIS groups and Steiner systems**  
with C.F. Nyberg-Brodda - in preparation
- [3] **Minimal relations of relational structures**  
with S.D. Freedman and C.M. Roney Dougal - in preparation
- [4] **Imprimitive maximal cocliques in the generating graphs of  $A_n$  and  $S_n$**   
with C.M. Roney-Dougal - in preparation
- [5] **On String C-Group Representations of Covers of the Sporadic Simple Groups**  
with P.J. Rowley - in preparation

## Invited Visits

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|------|-------------------------------------------------------------|------------------------------------------------------|
| 2025 | <b>Korea Institute for Advanced Study</b>                   | Visiting scholar and speaker, 2 weeks                |
| 2023 | <b>CMI-HIMR Summer School in Probabilistic Group Theory</b> | Teaching assistant and speaker                       |
| 2022 | <b>Young Group theorists workshop</b>                       | Participant and speaker at SwissMap Research Station |
| 2022 | <b>MFO Research Visitor</b>                                 | Research collaboration in Oberwolfach, 2 weeks       |
| 2022 | <b>INI Research Visitor</b>                                 | Research visit and workshop attendee, 2 weeks        |
| 2020 | <b>Research in “Pairs”</b>                                  | Research collaboration at Oberwolfach, 2 weeks       |
| 2020 | <b>INI Research Visitor</b>                                 | Research visit and workshop attendee, 2 weeks        |

## Awards and Funding

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|---------|----------------------------------------------------------|------------------------------------------------------------------|
| 2023-25 | <b>HIMR small grant £1840 p.a.</b>                       | Funding for a research seminar                                   |
| 2024    | <b>STEM for Britain finalist</b>                         | Poster competition and exhibition held in Parliament             |
| 2023    | <b>HIMR small research grant £1000</b>                   | Funding to host a collaborator for 1 week                        |
| 2023    | <b>MIMS funding £700</b>                                 | Funding to host a collaborator for 1 week                        |
| 2018    | <b>St Andrews PhD scholarship</b>                        | Fees and stipend for 3 years                                     |
| 2018    | <b>Outstanding Academic Achievement Award £1000</b>      | Top 0.5% of Manchester students                                  |
| 2018    | <b>IMA Prize 2018</b>                                    | Final year outstanding project award                             |
| 2018    | <b>Manchester President’s Doctoral Scholar Award</b>     | Fees and stipend for 3.5 years (declined)                        |
| 2018    | <b>Nominated for The Distinguished Achievement Award</b> | Undergraduate of the Year                                        |
| 2017    | <b>Fourth Year Scholarship</b>                           | Tuition fee waiver                                               |
| 2017    | <b>LMS Undergraduate Research Bursary £1440</b>          | Funding for a summer research project                            |
| 2016    | <b>BP achievement award £2000</b>                        | For an essay on Big Data                                         |
| 2015-6  | <b>Dalton scholarship £300</b>                           | Academic achievement in 1 <sup>st</sup> and 2 <sup>nd</sup> year |

## Invited Talks

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| Jul 2026 | <b>Groups arising from puzzles and games</b>                           | Postgraduate Group Theory Conference                                                |
| Feb 2026 | <b>Matroids and IBIS groups</b>                                        | Bristol Algebra Seminar                                                             |
| Oct 2025 | <b>Irredundant bases for permutation groups</b>                        | Korean Institute of Advanced Study, Combinatorics And Representation Theory Seminar |
| Sep 2024 | <b>Base and irredundant base size - how different are they really?</b> | Topics in Group Theory - on the occasion of Lucchini’s 60th birthday, Padova        |
| Sep 2024 | <b>Cliques and cocliques in generating graphs</b>                      | Dualities in Prob. and Alg., Lancaster                                              |
| Jun 2024 | <b>My favourite group</b>                                              | British Early Career Mathematicians’ Colloquium, Birmingham                         |
| Nov 2023 | <b>Generating groups</b>                                               | Birmingham Algebra Seminar                                                          |
| Oct 2023 | <b>Relational complexity of linear groups</b>                          | Warwick Algebra Seminar                                                             |
| Mar 2023 | <b>Calculating relational complexity</b>                               | London Algebra Colloquium                                                           |
| Sep 2022 | <b>Generating graphs and feeling good enough</b>                       | Piscopia Society PiWorks seminar                                                    |
| Sep 2022 | <b>A survey of base size</b>                                           | Young Group Theorists Workshop                                                      |
| Feb 2022 | <b>Relational complexity</b>                                           | St Andrews Pure Colloquium                                                          |
| Nov 2021 | <b>Group statistics</b>                                                | Bristol Junior Algebra Colloquium                                                   |
| Nov 2021 | <b>Maximal irredundant base size of linear groups</b>                  | Birmingham Algebra Seminar                                                          |
| Jan 2021 | <b>Relational complexity of permutation groups</b>                     | Functor Categories meeting                                                          |
| Oct 2020 | <b>Base size of permutation groups</b>                                 | North British Semigroups Network                                                    |
| Jun 2020 | <b><math>M_{24}</math> and octad triples</b>                           | Junior London Algebra Colloquium                                                    |
| May 2020 | <b>Maximal subgroups and maximal cocliques</b>                         | York Algebra Seminar                                                                |

## Teaching

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- 2025 **Galois Theory** Co-leading a level 4 course with 22 students  
Personal responsibilities: 12 lectures, 6 tutorials, midterm and exam marking
- 2025 **Coding Theory** Wrote and delivered a 12 hour course for 10 postdocs
- 2023 **O’Nan Scott Theorem** Lead a reading group for 8 people
- 2015-21 **Tutorial lead** Various undergraduate modules at Manchester & St Andrews University  
Personal responsibilities: lead tutorials and marked work for approx 15 students per class

## Outreach, EDIA and Mentoring

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- 2024-25 **Manchester EDIA Forum** Built a postdoc EDIA webpage with resources and created a poster for Black History Month
- 2024-25 **HIMR EDIA Working group** Volunteered at a pre interview event, helped create a culture survey, and collated advice on giving accessible talks
- 2023-25 **LMS Levelling Up** Ran weekly tutorials and created revision resources for five A-level students from under-represented backgrounds
- 2025 **Inclusivity in Mathematical sciences** Participated in two workshops focusing on creating resources which highlight the work of non-western mathematicians and promote EDIA
- 2025 **EDIA “reading” group** Founded a reading group focusing on more niche EDIA topics with a book, podcast and film for each session to ensure accessibility  
Previous: Freedom House EMS, 504 sit-in. Upcoming: social housing design, Welsh Not
- 2018-23 **Mentor** Volunteered as a mentor in peer mentoring schemes for undergraduate and masters students at the universities of Manchester and St Andrews
- 2024 **Mathematical Modelling and Problem Solving** Lead a workshop on cryptography for local secondary school students
- 2024 **MAP supervisor** Created and supervised two projects for A-level students on the Manchester Access Programme, a widening participation scheme
- 2024 **Martingale Foundation** Academic interviewer for scholarships for individuals from low-socioeconomic backgrounds
- 2023 **Women in Maths Day** Created and ran a interactive workshop session on mazes for GCSE and A-level students

## Other Activities

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- Coding** Experienced in MAGMA and GAP
- British Sign Language** Obtained Level 2, studying for Level 3
- Ultra running** 50 mile and 50k races, started an instagram community with over 30k followers
- Braille** Learning level 1

## References

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|---------------------------------------------------------|-------------------------------------|
| <b>Prof Alexandre Borovik</b> MMath Dissertation Marker | alexandre@borovik.net               |
| <b>Prof Sarah Hart</b> Collaborator                     | s.hart@bbk.ac.uk                    |
| <b>Prof Colva M. Roney-Dougal</b> PhD Supervisor        | colva.roney-dougal@st-andrews.ac.uk |